

A Physics-AI Digital Twin of an Actuated Heated Channel: CFD Reference, Neural Surrogates, Sparse-Sensor Assimilation and Control with PINNeAPLe

Can a fast surrogate replace transient CFD for actuator trajectories it has never seen?

PINNeAPLe Labs
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Abstract

We implement, end to end, the four-layer digital-twin study of a 2-D channel with a heated wall section driven by a time-varying inlet velocity $U_{in}(t)$ and heater power $Q_h(t)$. (1) Computational physics: a transient incompressible Navier–Stokes + energy solver added to PINNeAPLe, verified against plane Poiseuille flow (second-order convergence of the profile), the fully developed Nusselt number for one-sided uniform heating ($Nu = 5.388$ vs 5.385 , error 0.06%) and a three-grid convergence study; 82 CFD trajectories with random, chirp and edge-step actuations. (2) Scientific ML: POD + DMDc, POD + MLP latent dynamics, a convolutional autoencoder, FNO and DeepONet from PINNeAPLe, and a physics-informed FNO with continuity and global energy-balance losses, all rolled out autoregressively for 100 steps. The model selected on validation (POD + MLP latent dynamics) reproduces the temperature field of unseen chirp actuations with relative L2 error 0.093 and outlet temperature RMSE 0.0037 , roughly $34\times$ faster than the CFD. The physics-informed FNO improved on the data-only FNO (0.336 vs 0.393) but neither approached POD + MLP; POD + DMDc diverged. (3) Data assimilation: with three noisy sensors, an ensemble Kalman filter (PINNeAPLe) changed the mean field error only marginally (miscalibrated heater: $0.221 \rightarrow 0.197$ with joint gain estimation), and gain estimation degraded the calibrated cases ($0.059 \rightarrow 0.195$). (4) Control: a heater schedule optimised through the differentiable surrogate and re-simulated in CFD tracked the target with RMSE 0.0340 , against 0.0331 predicted by the surrogate.

Keywords: digital twin; CFD; neural operators; FNO; DeepONet; POD; data assimilation; EnKF; model predictive control; PINNeAPLe

1 Layer 1 — CFD reference

Non-dimensional channel $[0, 10] \times [0, 1]$, $Re = 100$ (based on height and reference inlet velocity), $Pr = 0.71$, heater on the bottom wall for $2 \leq x \leq 4$ with prescribed flux $Q_h(t)$, adiabatic elsewhere, parabolic inlet scaled by $U_{in}(t) \in [0.6, 1.4]$, $Q_h \in [0.0, 2.0]$. Staggered MAC grid 128×32 , donor-cell/central convection blend, Chorin projection with a pre-factorised pressure Poisson matrix, $\Delta t = 0.01$; outlet mass correction (PINNeAPLe *ThermalChannel2D*).

Table 1. Verification of the CFD reference. The GCI study shows oscillatory convergence for the transient QoIs (differences of a few percent between grids), which bounds the accuracy the surrogates can meaningfully be held to.

Check	Result
Poiseuille profile error, $64 \times 16 / 128 \times 32 / 256 \times 64$	$4.8e-03 / 1.3e-03 / 3.5e-04$
dp/dx vs exact $12/Re$ (finest grid)	0.1187 vs 0.1200
max l_{div} ul	$2.6e-07$
Nusselt (fully developed), $128 \times 16 / 256 \times 32$	$5.392 / 5.388$ (reference 5.385)
global energy imbalance at steady state	$2.7e-08$
GCI (fine) of T_{out}	0.015 , observed order 1.34
GCI (fine) of T_{max}	0.002 , observed order 2.28
GCI (fine) of dp	0.005 , observed order 1.28

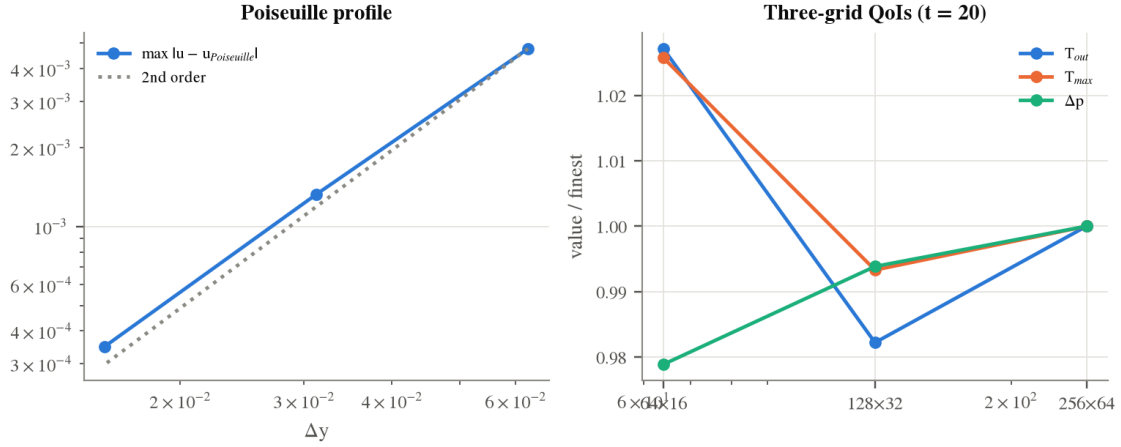


Figure 1. Left: second-order convergence of the Poiseuille profile. Right: QoIs of a transient scenario on three grids.

The dp/dx value is 1.1% below $12/Re$ even on the finest grid; this is consistent with the first-order ghost-cell treatment of the no-slip wall and is reported rather than tuned away.

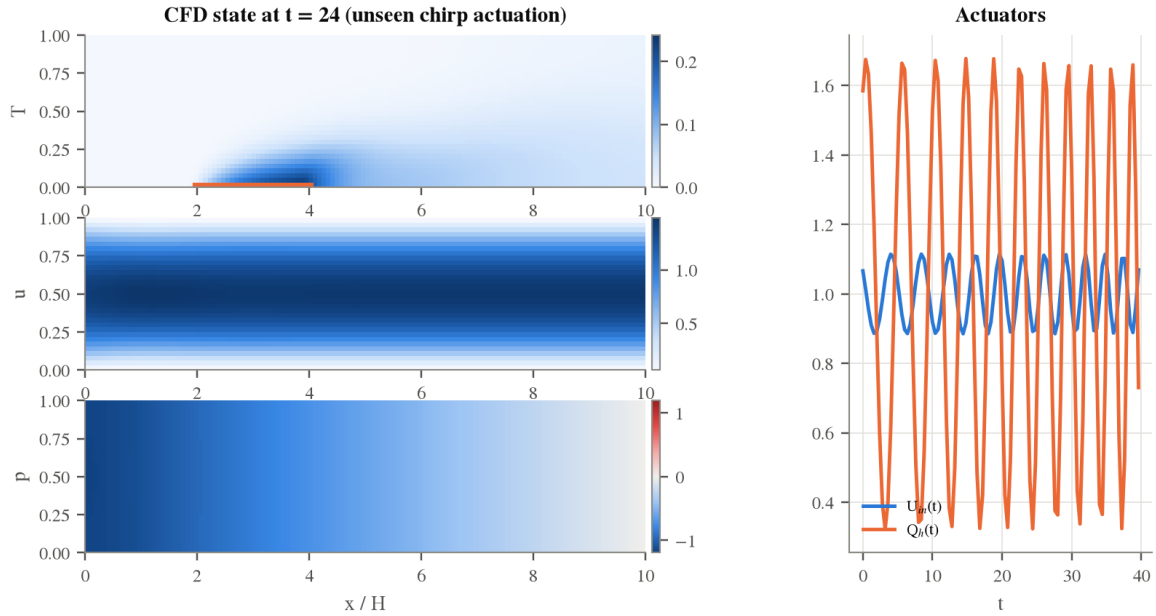


Figure 2. One CFD state (temperature, streamwise velocity, pressure) during an unseen chirp actuation; orange bar: heater.

Dataset: 82 trajectories, each 15 time units of warm-up followed by 40 recorded units at $\Delta t = 0.4$ (101 snapshots): 48 training and 8 validation trajectories with random step/ramp actuations; 12 test trajectories of the same family; 8 with chirp (frequency-sweeping sinusoidal) actuations and 6 with steps between the range edges — two families never seen in training. Mean CFD cost 42 s per trajectory (single core).

2 Layer 2 — Surrogates

- POD (training snapshots, 99.99% energy, ≤ 40 modes per field; 40 modes in total) + DMDC (linear latent map with actuator input).
- POD + MLP latent dynamics trained on 8-step latent rollouts.
- Convolutional autoencoder (latent 64) + latent MLP stepper; field loss over 4-step rollouts.
- FNO: PINNeAPPLE FNO2d with zero padding (non-periodic channel), actuator samples as channels, residual update, 4-step rollout loss.
- Physics-informed FNO: $+\lambda[\text{continuity}^2 + \text{global-energy-balance}^2]$ computed on the de-normalised predictions; $\lambda \in \{0.1, 1\}$ chosen on validation.
- DeepONet (PINNeAPPLE): branch = POD coefficients + actuator samples, trunk = (x, y) .

- All models: training statistics only; selection and early stopping on validation rollouts; test families scored once. Field models (CAE, FNO, PI-FNO): up to 20 epochs, patience 5, rollout windows starting at every third snapshot (random offset per epoch) — a budget imposed by the shared laptop GPU, identical for all three.

Table 2. Rollout metrics over 100 steps (40 time units). Energy residual: global energy-balance mismatch per step, relative to the maximum heat input; the CFD data themselves give 0.008 (discretisation floor of the diagnostic).

Model	val. rollout rel. L2	T rel. L2 (same family)	T rel. L2 (unseen: chirps)	T rel. L2 (unseen: edge steps)	T_{out} RMSE (chirps)	energy residual (chirps)	rollout s
POD + DMDc (linear latent)	diverged	diverged	diverged	diverged	—	—	0.04
POD + MLP latent dynamics (selected)	0.335	0.044	0.093	0.086	0.0037	0.163	1.23
Conv autoencoder + latent MLP	0.364	0.093	0.141	0.383	0.0039	0.414	1.46
FNO (PINNeAPLe FNO2d)	0.342	0.136	0.393	0.314	0.0065	0.150	3.06
Physics-informed FNO	0.341	0.117	0.336	0.259	0.0054	0.126	131.72
DeepONet (PINNeAPLe)	0.988	0.840	0.831	0.922	0.0251	2.604	2.76

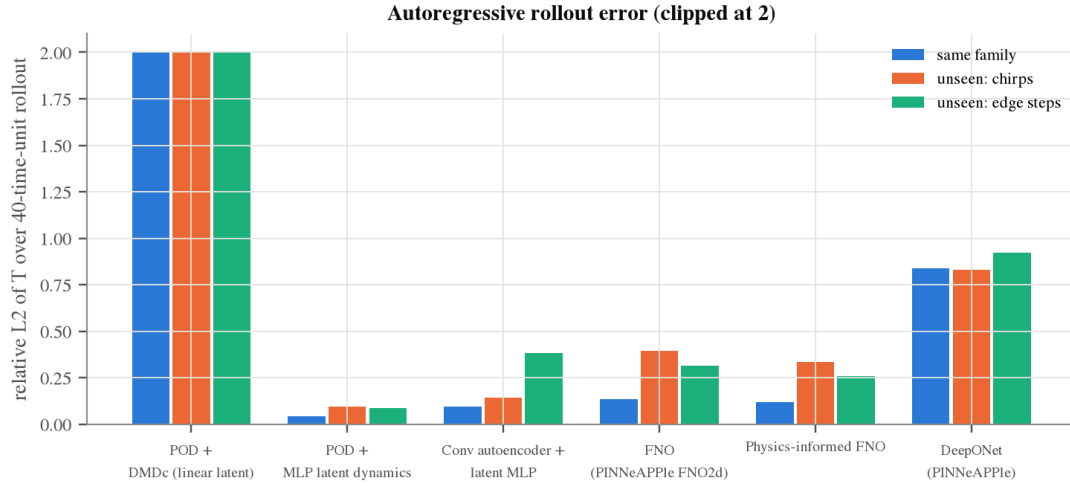


Figure 3. Relative L2 error of the temperature field over the full rollout, by model and test family.

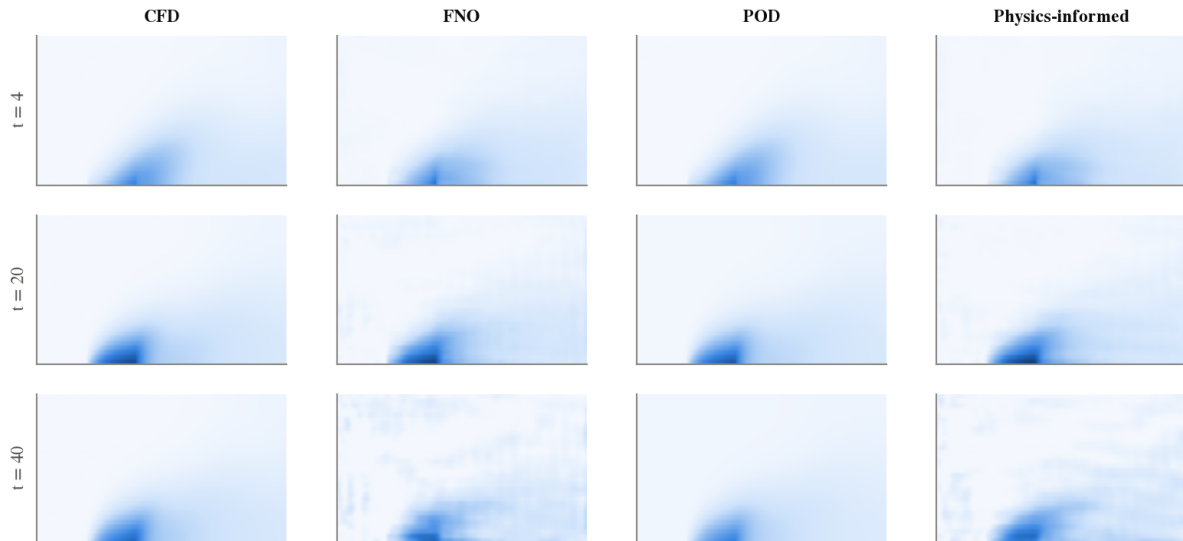


Figure 4. Temperature field of one unseen chirp trajectory: CFD vs surrogate rollouts at three times.

Physics-informed vs data-only FNO ($\lambda = 0.1$): temperature error on unseen chirps 0.336 vs 0.393; energy residual 0.126 vs 0.150. The physics loss helped the FNO on both counts, but both FNO variants remain well behind the selected POD + MLP model (0.093). Rollout times were measured on a laptop shared with other jobs: the two FNOs have identical architectures, so the difference in their rollout times reflects machine load at the moment of measurement, not the models.

3 Layer 3 — Sparse sensors and state correction

Table 3. Relative L2 error of the temperature field with three sensors T1, T2, P1 (T at (5, 0.5) and (9, 0.25), p at (0.5, 0.5); noise $\sigma = 0.002$ for T, 0.02 for p). The 'plant' is the CFD with the heater delivering gain $\times$ the commanded power (1.0 = nominal, 1.3 = uncalibrated heater); the surrogate only knows the command. EnKF: PINNeAPPLE EnsembleKalmanFilter, 48 members, on the POD + MLP latent state.

Run	family	true gain	open loop	EnKF state	EnKF state + gain	sensors only	gain estimate
56	same family	1.0	0.038	0.032	0.048	0.825	1.003
56	same family	1.3	0.200	0.179	0.065	0.847	1.263
57	same family	1.0	0.029	0.033	0.086	0.841	0.970
57	same family	1.3	0.215	0.206	0.176	0.858	1.255
58	same family	1.0	0.047	0.036	0.080	0.863	0.779
58	same family	1.3	0.210	0.190	0.079	0.876	1.061
59	same family	1.0	0.042	0.049	0.213	0.847	0.964
59	same family	1.3	0.214	0.188	0.240	0.860	1.063
68	unseen sinusoid	1.0	0.074	0.079	0.205	0.819	0.687
68	unseen sinusoid	1.3	0.237	0.213	0.216	0.832	1.015
69	unseen sinusoid	1.0	0.126	0.164	0.537	0.862	0.764
69	unseen sinusoid	1.3	0.253	0.292	0.404	0.864	1.217

Averaged over the 6 cases with a calibrated heater (gain 1.0): open loop 0.059, EnKF on the state 0.065, EnKF with gain estimation 0.195. With the uncalibrated heater (gain 1.3): 0.221, 0.211 and 0.197; the estimated gain ends between 1.01 and 1.26. Three sensors carry little information about the full temperature field: the sensor-only reconstruction fails (error ≈ 0.850), and the filter corrections are small. Estimating the gain helps somewhat when the heater really is miscalibrated but harms the calibrated cases, where the estimate drifts below 1 (down to 0.69); a gain prior tighter than the one used, or more sensors near the heater, would be needed before relying on it.

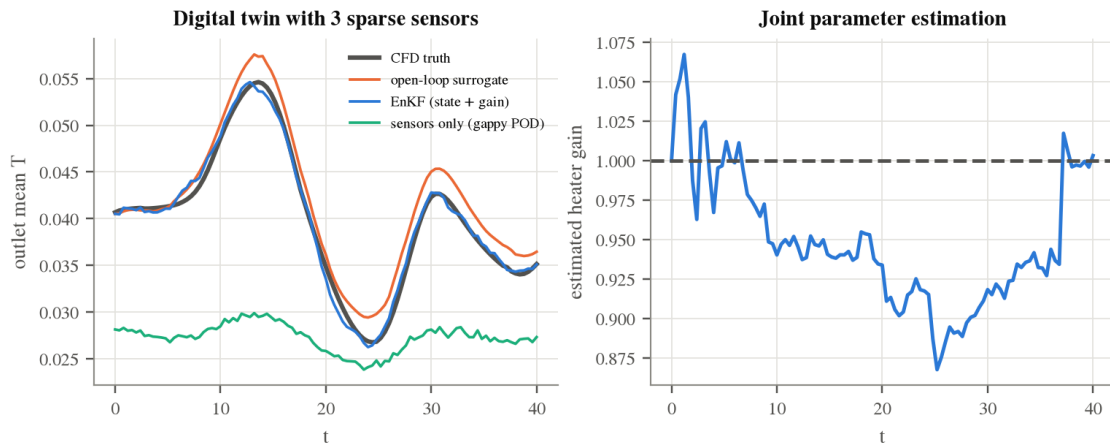


Figure 5. Left: outlet temperature for one case — CFD, open-loop surrogate, EnKF-corrected twin and sensor-only reconstruction. Right: the heater-gain estimate of the augmented EnKF.

4 Layer 4 — Optimisation verified in CFD

Goal: hold the outlet temperature at 0.05 with $U_{in} = 1$ while penalising heater power, by optimising 10 piecewise-constant heater levels through the differentiable POD + MLP surrogate (722 s of Adam). The optimised actuation was then simulated with the CFD (15 s): tracking RMSE predicted by the surrogate 0.0331, obtained in CFD 0.0340; surrogate–CFD mismatch of T_{out} $9.72e-04$.

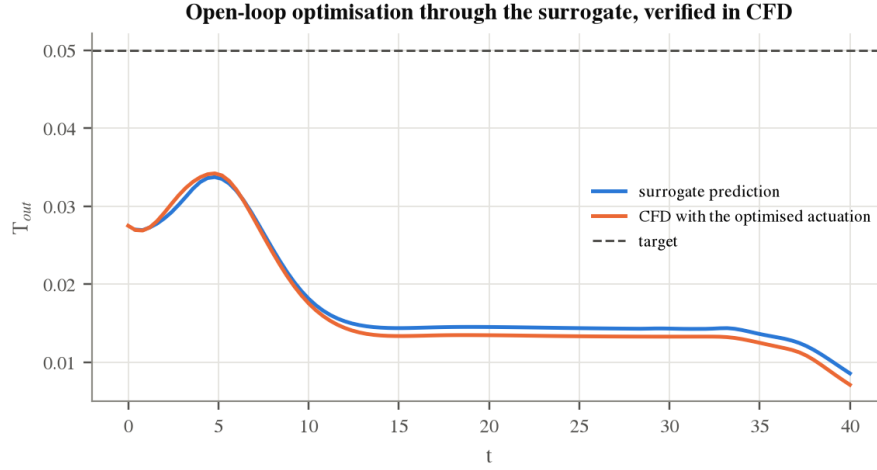


Figure 6. Outlet temperature under the optimised heater schedule: surrogate prediction vs the CFD re-simulation.

5 VeriPhysics validation

Table 4. numerical_convergence is the CFD reference's own 3-grid GCI on T_{out} (shared by all surrogates); physics_guardrail not run: no compiled ProblemSpec for this time-dependent actuated NS+energy system; uq_calibration not run: the surrogates are deterministic single models

Model	Trust score	Coverage	Components / note
POD + DMDc (linear latent)	—	0.0	rollout diverged (overflow in the error)
POD + MLP latent dynamics	0.925	0.4	numerical_convergence 1.00; benchmark_agreement 0.85
Conv autoencoder + latent MLP	0.929	0.4	numerical_convergence 1.00; benchmark_agreement 0.86
FNO (PINNeAPLe FNO2d)	0.869	0.4	numerical_convergence 1.00; benchmark_agreement 0.74
Physics-informed FNO	0.896	0.4	numerical_convergence 1.00; benchmark_agreement 0.79
DeepONet (PINNeAPLe)	0.546	0.4	numerical_convergence 1.00; benchmark_agreement 0.09

6 Answer to the project question

Can a CFD run be replaced by a fast Physics-AI representation for interactive evaluation inside a known operating envelope? For quantities of interest, largely yes: the selected POD + MLP model predicts the outlet temperature of unseen actuation families with RMSE 0.0037 (chirps) about 34× faster than the CFD, and a heater schedule optimised through it achieved in CFD a tracking error of 0.0340 against the 0.0331 it predicted. For full fields, only partly: the temperature-field error roughly doubles from the training family to unseen families ($0.044 \rightarrow 0.093$ and 0.086), and every surrogate violates the global energy balance by far more than the CFD's own diagnostic floor (0.008). Sparse-sensor assimilation with three sensors did not materially correct the surrogate; the digital-twin layer is the weakest link of this study.

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